



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Project Task 2

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Section: 02

Group name: Router ranger

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Question

1. What are the specific expectations for wireless connectivity and range coverage across both floors?

The wireless connectivity must provide uninterrupted access to the building, including all labs, student lounges, and common areas. Cisco Network Lab and Hybrid Classroom will host bandwidth-intensive activities such as online conferencing and real-time collaboration, it required a high bandwidth and speed of Wi-Fi with minimal. Therefore, the Wi-Fi routers should place at least one per room because to eliminate dead zones problem and ensure reliable connectivity for all users. These routers should be chosen to accommodate high-density usage, especially in labs where multiple devices connect simultaneously.

2. What devices and functionalities are required for the Cisco Network Lab?

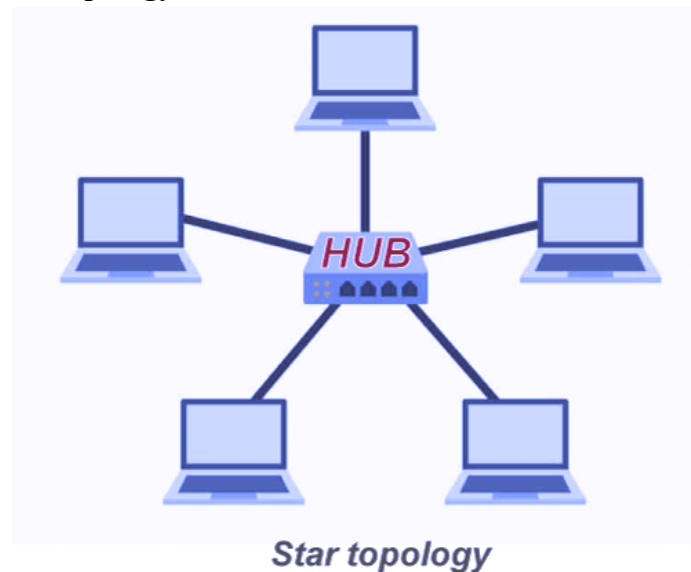
For an ideal setup in the Cisco Network Lab, the recommended equipment is two cisco catalyst 3750 switches [1]. These switches provide Layer 3 routing capabilities and are excellent for teaching network switching and routing protocols. They support VLAN configurations, enabling students to learn network segmentation and traffic management in a controlled environment. Another recommended equipment is cisco 2800 series routers [1]. These routers support basic routing protocols and WAN connections which are vital for understanding and configuring network communication across different networks. The Cisco 2800 routers also support multiple interfaces which allow students to connect different network segments.

3. What specific security protocols should be implemented to protect against common threats like DDoS attacks and internet worms?

First, firewalls should be deployed at key network entry points to monitor and control incoming and outgoing traffic based on predetermined security rules [2]. Intrusion Detection and Prevention Systems (IDPS) are also essential, as they both passively monitor traffic and actively block suspicious or malicious behaviour once flagged. Second, network segmentation divides the network into secure segments or

VLANs based on different user roles, such as student and staff access [3]. This limits access within the network and minimizes the impact of a breach in one segment from affecting the entire network. Third, we can implement the 802.1X Authentication combined with Radius Server [4]. This protocol enforces network access control by requiring users to authenticate with UTM student ID and password before connecting to the Wi-Fi. It can secure the student privacy since 802.1X authentication protects people to walk in and release virus, malware or man-in-the-middle attack.

4. What network topology would be suitable inside the labs?



The network topology used varies in many aspects, especially since the faculty building contains two floors and several different labs. In our group opinion, the most suitable network topology for the labs in this project is the Star Topology. In a star topology, all devices such as workstation and IoT devices are connected to a central hub or switch [5]. This central hub allows for easier network management, as problem can be isolated and resolved quickly without affecting the entire lab network. If one connection fails, it does not affect the others, making it more reliable.

Furthermore, each device has a direct link to the switch, there is minimal network congestion which resulting lower latency and faster data transmission speeds. This is crucial for lab environments, especially the Cisco Network Lab and Embedded Lab, where high-speed internet and seamless connectivity are required for educational activities.

5. What is the minimum bandwidth for each room?

There are a total of 4 labs, namely 2 general-purpose lab, 1 Cisco Network lab and 1 Embedded lab. We plan to install an internet router with a minimum 30Mbps network speed per device. This is to ensure overall smooth experience in preparation for education in line with 4IR, which includes IoT, AI-driven applications, big data, and real-time analytics. We also consider the large number of students (30 students per lab) when assessing suitable internet speeds, as high network usage from activities like streaming and resource-intensive applications (e.g., Cisco Webex meetings) can place significant demands on bandwidth. This totals up to a minimum 1Gbps per lab.

Besides, the video conference room needs a higher internet bandwidth. Since it is only equipped with a Smart TV for online meeting purposes, it will be connected through ethernet to the router for a more stable connection. Additionally, this setup minimizes interference and enhances security by creating a more direct, controlled connection to the network. Besides, taking into consideration that the streaming quality and audio quality should be top notch, we plan to install a minimum 100Mbps dedicated router for the room.

Moreover, there is a hybrid classroom which require simultaneous high-quality video streaming and interactive platforms (e.g., real-time collaboration tools). The video streaming platform should provide high image and audio quality. Next taking the total number of students into account, the network would need to handle multiple connected devices. In conclusion, with 30 Mbps provided for each device, the total bandwidth for hybrid classroom should be 1Gbps.

The new building also includes new facilities such as student lounge and a student meeting room. We assume that both rooms are equipped for students to do normal daily tasks such as web surfing and online meeting, hence a minimum 10 Mbps should suffice for each device. Since there could be lots of students staying in both rooms, the total bandwidth should be increased to provide adequate speeds that can support multiple users. There's also a working office which can accommodate up

to 8 lecturers. If lecturer also handle web browsing and online meetings, a minimum 3 Mbps should suffice for each device too. To conclude, a total of 100Mbps for student lounge and student meeting room each, and total of 24Mbps for the office.

Lastly, the new building is also equipped with a server room. It's a small-scale server room used to support basic file storage, internal applications, and email servers. With that in mind the bandwidth speed should be at least 100Mbps. This ensures smooth performance for daily tasks and prevents congestion, while also allowing for future scalability as the network demands grow.

6. What type of networking cable is suitable to connect to the computers? Why did you choose it?

Category 6 Ethernet cable would be a suitable choice for connecting the computers. Category 6 cables are a type of network cable used for connecting devices to a local area network (LAN) because they offer a good balance between speed, distance, and cost [6]. With a bandwidth capacity of up to 250 MHz, Category 6 cables can support data transfer rates of up to 1 Gbps (Gigabit Ethernet) over distances up to 100 meters, which is ideal for most classroom, office, and lab environments shown in our design [7]. This makes it appropriate for handling the data requirements of labs, classrooms, and conference rooms, where multiple devices may need to transfer large amounts of data simultaneously. We choose this cable as it fits all the needs for each machine to communicate and connect to the Internet seamlessly.

The structured layout of our floor plans suggests spaces where a stable and high-speed network connection is essential, such as in the Cisco Network Lab, General Labs, and the Hybrid Classroom. Category 6 cables are designed to reduce crosstalk and interference, which is especially important in high-density environments where numerous cables may run close to each other [7]. This cable type will ensure a reliable connection for video conferencing, streaming, and network-dependent software used in collaborative spaces, like the video conference room and the student lounge area.

In addition, using Category 6 cables allows for future scalability. If network demands increase, Cat6 can support 10 Gigabit Ethernet over shorter distances (up to 55 meters), which could be beneficial if Faculty Computing plans to upgrade to higher-speed networks. For areas with high traffic, such as the server room or labs, this extra capacity could be valuable. Overall, Category 6 Ethernet cables offer the best combination of performance, cost-effectiveness, and futureproofing, making them a suitable choice for the networking needs reflected in our design.

7. How to ensure that the coverage of wireless networks can support the area of the faculty?

In our opinion, there are several ways to ensure that the coverage of wireless networks can support the area of the faculty. First, choosing a central location at each floor for the router can improve the coverage of wireless networks as a router in the central position provides the perfect signal coverage for the entire building.

Besides, adding a wireless repeater is also a good way to ensure the coverage of wireless networks. Wireless repeater can rebroadcast a wireless signal and strengthen the signal from the router to other floors of the building. We can place the wireless repeaters at the classrooms or labs that receives weak wireless signals so that the performance of the network in those rooms can be improved.

Apart from that, we can also use an ethernet cable to connect the access point to the router by inserting the cable into a LAN port on the router. This introduces the access point to the router's created local area network then connects the access point to the computer with an ethernet cable so that the computer can direct access to access point and LAN management. Once the wireless access point is done, students can enjoy expanded wireless coverage and improved wireless signal. The issues that could arise when dealing with coverage is that the maximum number of clients that can be served by a single access point highly depends on the fact that what internet services are being availed by the users. Then, sometimes users can associate with an access point located in a distance area instead of connecting to a nearby access point which is offering weak signal strength due to co-channel interference or interference caused by walls or other solid substances [5].

8. Is there a preference for wired or wireless connectivity in different areas of the building, such as labs versus the student lounge?

Yes, there is a distinct preference for different types of connectivity based on the specific requirements of each area in the building. In the general-purpose labs, Cisco Network Lab, and Embedded Lab, wired connectivity is preferred. This is because wired connections provide more stable, high-speed, and low-latency internet access, which is essential for activities like running network simulations, IoT experiments and high-bandwidth applications. Wired connections also offer better security and reliability compared to wireless, making them ideal for labs where consistent performance is crucial.

In areas like the student lounge, meeting rooms, and cafe, wireless connectivity is preferred. These are spaces where flexibility and mobility are prioritized, allowing students and staff to connect their laptops, smartphones, and tablets without being constrained by physical cables. Wi-Fi access points provide convenient, high-speed internet access for users to study, collaborate on projects, or relax while staying connected.

9. What are the key elements in a hybrid classroom?

In a hybrid classroom, a reliable high-speed internet connection is the foundational network element that ensures seamless connectivity for both in-person and remote participants. This robust connection supports online resources, live video conferencing, and interactive platforms essential for hybrid learning. Since multiple users, both in the classroom and online, will be accessing resources simultaneously, a strong internet connection prevents lag or disruptions and enables real-time collaboration. Typically, a wired or fibre-optic internet link is preferred for its stability and faster speeds compared to wireless options.

Wireless Access Points (WAPs) are crucial in a hybrid classroom to provide Wi-Fi coverage throughout the room. While wired connections are preferred for

devices that require constant, high-speed connectivity, WAPs enable students and instructors to connect laptops, tablets, or other devices wirelessly. A strong Wi-Fi signal ensures flexibility in seating arrangements and smooth access to online resources, facilitating effective participation for in-person students while interacting with remote learners.

Ethernet cabling, such as Category 6 (Cat6) Ethernet cables, is essential for providing stable, high-speed connections to key devices like the instructor's computer, streaming equipment, and interactive displays. These wired connections reduce latency and improve performance for high-bandwidth activities like live streaming and video conferencing. A wired setup minimizes the chances of connectivity issues that may affect the quality of audio and video feeds during hybrid sessions, making it an essential component of a reliable hybrid classroom.

A network switch connects multiple devices within the classroom to the local area network (LAN) and efficiently manages network traffic between them. The switch links devices such as computers, streaming equipment, displays, and possibly Internet of Things (IoT) devices, enabling seamless communication across the network. By serving as a central point for network management, the switch prevents bottlenecks and ensures that each device receives the necessary bandwidth, even during periods of high network usage.

Finally, firewall and security solutions are critical to safeguard the hybrid classroom network. With multiple devices and participants connected, strong cybersecurity measures prevent unauthorized access and protect sensitive data. A firewall monitors and filters incoming and outgoing network traffic, ensuring data privacy and network safety. This secure environment allows instructors and students to focus on learning without concerns about potential cybersecurity threats.

10. How do you plan to reduce Internet service downtime issues?

We acknowledge that network downtime poses a significant threat as it disrupts the flow of the class and online meetings, leaving multiple users with an unsatisfactory learning environment. First, we plan to install a backup router and

network switches to deal with primary device failure. These can be automatically switched over without affecting the network, thus ensuring no interruption to online activities.

Besides, we can prevent downtime issue by installing multiple ISPs to the new building [8]. These ensure that whenever one ISP encounters slow speed or outage, the other ISP will immediately take over, hence reducing network delay in the process. Both measures utilise the server room that aim to solve the problem from the aspect of network redundancy, making it more cost effective.

For extra protective measure, we plan to install proactive network monitoring tools such as SolarWinds, Auvik and Datadog. These tools are tasked to detect any network traffic irregularities and connectivity issues, hence giving IT staff more time to solve the issues.[9]

11. What budget constraints should be considered when choosing networking devices and equipment?

When selecting networking devices and equipment for the project, several key budget constraints need to be considered, starting with the initial equipment costs. The primary expenses include the cost of networking devices such as workstations, switches, routers, access points, firewalls, and IoT devices for the Embedded Lab and Cisco Network Lab.

In addition, the building needs to accommodate 1,940 people, including students and staff, with an expected 15% growth over the next 4 years. Given this projected increase, it is crucial to select high-performance devices that can support future needs with cutting-edge technology, without requiring a complete overhaul. Although scalable devices may have higher upfront costs, they can save money in the long run by reducing the need for frequent upgrades.

Lastly, maintenance and support costs must also be considered within the budget. The total cost of network devices includes not only the initial purchase but also ongoing maintenance, software updates, and technical support. It is more preferable to choose devices with extended warranty coverage and reliable customer service,

which can help minimize unexpected expenses. Additionally, premium brands often come with higher support costs, which should be factored into the overall budget.

Feasibility Assessment

After answering all the questions, we have gained a clear understanding of the project requirements. We have evaluated the feasibility assessment from three key aspects: economic feasibility, legal feasibility, and technical feasibility.

Economic feasibility

The project's original budget includes expenses for networking equipment such as workstations, switches, routers, access points, and cables. Since the number of students and employees is predicted to increase by 15% over the next five years, equipment selection needs to find a balance between cost-effectiveness and performance requirements in order to maintain long-term financial viability. Although they may cost higher up front, high-performance devices like Cisco switches provide more scalability and dependability, which reduces the need for regular upgrades. Besides that, we can select devices with extended warranty coverage which will help reduce unexpected expenses, maintenance, technical support and software upgrades. Additionally, the project should include a contingency budget for unexpected expenses, such as additional equipment needs or network upgrades, ensuring the project remains cost effective even if updates are necessary.

Technical feasibility

Technical feasibility studies ensure we have the technical resources needed to meet project requirements and turn ideas into results. This study confirms that required hardware and software components can be sourced affordably from reliable vendors like Huawei and Cisco, with official purchase options. Subscription-based software, such as Microsoft and antivirus programs, can be negotiated with providers. These hardware and software will ensure our technology are able to keep up with the current building development, making the process easier to handle. Additionally, a minimum of 5 IT staff members are required to manage equipment and server maintenance, ensuring the technical feasibility of the new building."

Legal feasibility

This is significant as a procedure to avoid any potential issues from a legal perspective. It is best to avoid legal rights and copyrights of the software used for the network equipment such as Microsoft Office and so on. Therefore, all the licensed software will be purchased through the legal pathway that we need to hand on to the Department of Digital Services (UTMDigital) to deal with. Furthermore, we also need to ensure the purchase of hardware is following the original and legal channel. To avoid legal issues, all hardware and software will be sourced from certified suppliers. UTMDigital will oversee licensed software, including UTM Webex and Microsoft products, ensuring compliance with regulations and preventing the use of pirated goods.

Meeting Minutes

Meeting Minutes #1

Date/Time		6/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Overviewing task 2 2. Suggest some question for task 2 3. Task distribution among group members	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	20 00		-
Neo Li Xin	20 00		-
Pravinraj a/l Sivabathi	20 00		-
Ahmad Ziyaad bin Mohd Abbas	20 00		-
Meeting details			
No.	Discussing item	Item details	Person-In charge
1	Overviewing the case study of task 2 and make a simple discussion	- Choh Jing Yi show the case study and rubric of the task 2 - Make a simple discussion of the project and make sure everyone group member understands the project requirements and goals	Choh Jing Yi
2	Suggestion for questions for task 2	-All members discussed and suggest question using help of google -Every person suggests 3 questions for the task	All group members
3	Choose good question and fulfil the task	-All members vote for desire question and select ten questions out of suggestion	All group members
4	Distribution task	-Choh Jing Yi distributed task equally -Each person does three question and complete one feasibility study	All group members
5	End of meeting	-We ended the meeting at 20.40 after the distribution task confirmed.	All group members

Meeting Minutes #2

Date/Time		8/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Discuss the appropriate answer for each question 2. Discuss the feasibility assessment 3. Write the report for meeting minute	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	20 00		-
Neo Li Xin	20 00		-
Pravinraj a/l Sivabathi	20 00		-
Ahmad Ziyaad bin Mohd Abbas	20 00		-
Meeting details			
No.	Discussing item	Item details	Person-In charge
1.	Explain the answer for each question	-Choh Jing Yi review the answer and find the best answer for the question	Choh Jing Yi
2.	Discuss the feasibility assessment	-Choh Jing Yi explain the feasibility assessment from each aspect and make sure the content is fullfill the aspect	Choh Jing Yi
3.	Write the report for meeting minute	-Choh Jing Yi write the report for the meeting minutes	Choh Jing Yi
4.	Meeting ended	-We ended the meeting at 21.30 after the review all the report and make a submission	All members

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